AP BIOLOGY TOPIC 8.5

Community Ecology: Species Interactions, Simpson's Index & Succession

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

ENE-4.A | ENE-4.B | SIMPSON'S DIVERSITY INDEX | NICHE PARTITIONING

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.5 (COMMUNITY ECOLOGY: SPECIES INTERACTIONS, SIMPSON'S INDEX & SUCCESSION)

CED Core Vocabulary

Ecological Community — All interacting populations of different species living in the same habitat.

2. Relative Abundance (Evenness) — How evenly individuals are distributed among those species (high diversity requires both high richness and high evenness so no single species dominates!).

Competitive Exclusion Principle (G.F. Gause) — Two species competing for the exact same limiting resources cannot stably coexist — the superior competitor drives the inferior one to local extinction (e.g.,...

Realized Niche — The actual narrower subset of its fundamental niche a species occupies in the presence of biotic competition (Balanus restricts Chthamalus to upper rocks).

Two Components of Species Diversity — 1. Species Richness: The total number of different species present in the community.

Simpson's Index — D_I = 1 − 0.38 = 0.62 !

Fundamental Niche — The full range of abiotic conditions and resources a species could potentially occupy in the absence of competitors (Chthamalus across entire intertidal zone).

Niche Partitioning & Character Displacement — Natural selection favors individuals that use different microhabitats, food sizes, or feeding times, reducing competition and causing sympatric species to...

§2. 1. COMMUNITY STRUCTURE & SIMPSON'S DIVERSITY INDEX

Richness + Evenness

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

1. Community Structure & Simpson's Diversity Index ➔ **Ecological Community** ➔ **Two Components of Species Diversity** ➔ **ALL**

- **Ecological Community:** All interacting populations of different species living in the same habitat.
- **Two Components of Species Diversity:**
  - **1. Species Richness:** The total **number of different species** present in the community.
  - **2. Relative Abundance (Evenness):** How evenly individuals are distributed among those species (high diversity requires *both* high richness and high evenness so no single species dominates!).
- **Simpson's Diversity Index (AP Formula Sheet):**

Diversity Index (D_I) = 1 − Σ (n / N)² (Range: 0 to 1; Higher value = Higher Diversity!)

- **n** = total number of organisms of a particular species | **N** = total number of organisms of **ALL** species combined.

[EXAMPLE & BIOLOGICAL APPLICATION] **Worked AP Simpson's Diversity Index Calculation:**

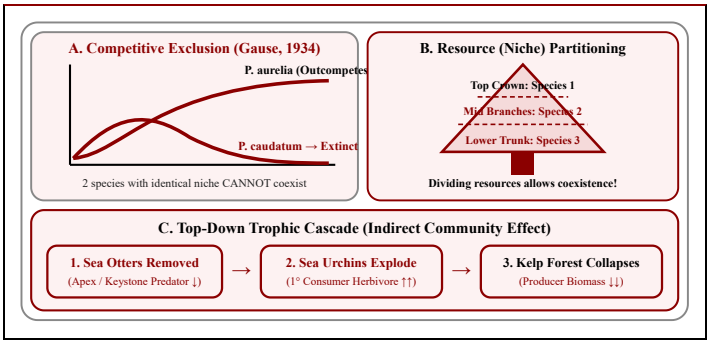
*A tide pool has 30 sea stars, 50 mussels, and 20 barnacles (Total **N = 30 + 50 + 20 = 100**):*

- **(n/N)²** for each species: Sea stars **(30/100)² = (0.30)² = 0.09**; Mussels **(50/100)² = (0.50)² = 0.25**; Barnacles **(20/100)² = (0.20)² = 0.04**.
- Sum of squares: **Σ(n/N)² = 0.09 + 0.25 + 0.04 = 0.38**.
- **Simpson's Index:** **D_I = 1 − 0.38 = 0.62** !

[COMMON AP EXAM FRQ TRAP] **High species richness alone does NOT guarantee high diversity!** Community A (25, 25, 25, 25 → **D_I = 0.75**) is far more diverse than Community B (97, 1, 1, 1 → **D_I = 0.06**) even though both have 4 species and 100 individuals!

§4. 3. COMPETITIVE EXCLUSION VS. NICHE PARTITIONING & CASCADES

Gause & MacArthur



- **Competitive Exclusion Principle (G.F. Gause):** Two species competing for the exact same limiting resources cannot stably coexist — the superior competitor drives the inferior one to local extinction (e.g., *Paramecium aurelia* outcompetes *P. caudatum* in mixed culture).
- **Fundamental vs. Realized Niche (Connell's Barnacles):**
  - **Fundamental Niche:** The full range of abiotic conditions and resources a species *could potentially* occupy in the absence of competitors (*Chthamalus* across entire intertidal zone).
  - **Realized Niche:** The actual narrower subset of its fundamental niche a species occupies in the presence of biotic competition (*Balanus* restricts *Chthamalus* to upper rocks).
- **Niche Partitioning & Character Displacement:** Natural selection favors individuals that use different microhabitats, food sizes, or feeding times, reducing competition and causing sympatric species to diverge morphologically (**character displacement**, e.g., Galápagos finch beak depths on shared islands).

§5. 4. ECOLOGICAL SUCCESSION & INTERMEDIATE DISTURBANCE

Community Change

- **Primary Succession (No Soil Present Initially — Slow):** Begins on lifeless, barren rock after lava flows, glacial retreat, or sand dunes. **Pioneer species (lichens & mosses)** weather rock and build the first organic soil → grasses → shrubs → climax forest.
- **Secondary Succession (Intact Soil Remains — Fast):** Begins after a disturbance (wildfire, flood, abandoned farmland, clear-cutting) where **nutrient-rich soil and seed banks remain intact**.
- **Intermediate Disturbance Hypothesis:** Moderate frequency/intensity of disturbance maximizes species diversity by preventing competitively dominant species from excluding everyone else while allowing colonizers to persist.

§3. 2. INTERSPECIFIC INTERACTIONS MATRIX (+/+, +/−, +/0, −/−)

Coevolution

INTERACTION	EFFECT	MECHANISM & CLASSIC AP EXAMPLES
Competition	− / −	Both species harmed competing for the same limiting resource. Drives **Competitive Exclusion** or **Resource (Niche) Partitioning**.
Predation / Herbivory	+ / −	Predator/herbivore benefits; prey/plant is harmed. Drives coevolutionary arms races: **cryptic camouflage**, **aposematic warning color**, **Batesian mimicry** (harmless mimics venomous, e.g., king snake mimics coral snake), **Müllerian mimicry** (two unpalatable species resemble each other, e.g., cuckoo bee & yellow jacket), and plant thorns/toxins (nicotine, tannins).
Mutualism *(Symbiosis)*	+ / +	Both species benefit: **Mycorrhizal fungi** (**P/H_2O** to plant roots ↔ plant sugars), ***Rhizobium*** bacteria in legume root nodules (**N_2** fixation), **Lichens** (fungus + algae), Coral + *Zooxanthellae*, Pollinators + Flowers.
Commensalism *(Symbiosis)*	+ / 0	One species benefits; the other is unaffected (e.g., barnacles on whales, cattle egrets eating insects flushed by grazing buffalo, epiphytic orchids on tree branches).
Parasitism *(Symbiosis)*	+ / −	Parasite derives nourishment from host (harming host without usually killing immediately): ticks/fleas (*ectoparasites*), tapeworms/*Plasmodium* (*endoparasites*).

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 8.5: Community Ecology: Species Interactions, Simpson's Index & Succession ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm² leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: $dN/dt = B - D$. Exponential Growth: $dN/dt = r_{max} \cdot N$ (J-curve). Logistic Growth: $dN/dt = r_{max} \cdot N \cdot [(K - N)/K]$ (S-curve; maximum instantaneous growth rate dN/dt occurs at $N = K/2$). Simpson's Diversity Index: $D_I = 1 - \Sigma(n / N)^2$ (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: $NPP = GPP - R$. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.5 to prove why the claim holds.

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